Solar Nuclear Astrophysics

Part I — Observations of the Sun's Core
Part II — Theory

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Primary sources: Borexino Collaboration, *Nature* **562**, 505 (2018) and *Nature* **587**, 577 (2020)

Theoretical sections after lectures by Prof. Anil Kumar Gourishetty, IIT Roorkee

How To Understand This Note

Part 1 and Part 2 answer different questions and can be taught in either order, though the sequence below (observation first, theory second) is the one that works best in the classroom

Part I (Observations)

Question: what has actually been measured about the inside of the Sun, and how?

Runs from the physics of neutrinos as a probe, through sixty years of detectors, to the two Borexino results: the complete pp-chain spectroscopy (2018) and the first CNO detection (2020)

Part II (Theory)

Question: what machinery do you need in order to say what those numbers *mean*?

Cross sections and S-factors, the Gamow peak, thermonuclear reaction rates, resonances, reaction networks, and the full chain from a core reaction rate to a count rate in a detector

Throughout, **blue** marks something that was *measured* and **red** marks something that was *deduced*. Keeping those two categories visually separate is the single most useful habit in this subject.

Outline

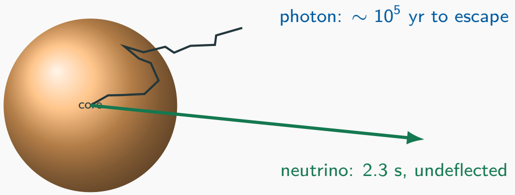
PART I

Observations of the Sun’s Core

I.1 Neutrinos, the Standard Solar Model, and the Detectors

The Sun Is Opaque to Its Own Light

A photon created in the solar core does not travel to us. It is scattered, absorbed and re-emitted continuously, random-walking through $\sim 7 \times 10^{10}$ cm of plasma with a mean free path of order a centimetre.



The consequence

Sunlight tells us the *total* power leaving the surface. It carries almost no information about *which nuclear reaction* released that power, and what information it does carry is $\sim 10^5$ years out of date.

What Makes the Neutrino the Decisive Probe

A neutral, nearly massless lepton emitted whenever a proton converts into a neutron by the weak interaction:

$$p \to n + e^+ + \nu_e \qquad \text{or} \qquad p + e^- \to n + \nu_e$$

- Interaction cross section $\sigma \sim 10^{-44}$–10^{-45} cm^2 at solar energies — some twenty orders of magnitude below a typical nuclear cross section.
- It therefore leaves the solar core without a single further interaction and arrives at Earth about eight minutes later carrying unmodified information about the reaction that produced it.
- The price of that transparency is symmetric: what is hard to stop is also hard to detect. Every experimental difficulty in Part I traces back to this one number.

What we actually measure

No detector observes a neutrino directly. All of them observe a *secondary* charged particle — an electron, or a radioactive daughter nucleus — produced when a neutrino happens to interact. Everything else is inference.

Every Weak Step Emits a Neutrino

The net transformation is the same for both hydrogen-burning routes, and it always emits two electron neutrinos:

$$4p + 2e^- \longrightarrow {}^4\mathrm{He} + 2\nu_e, \qquad Q = 26.73\ \mathrm{MeV}$$

pp chain $\sim 99\%$ of $L_\odot$

- $p + p \to d + e^+ + \nu_e$ continuum, ≤ 420 keV
- $p + e^- + p \to d + \nu_e$ line, 1.44 MeV
- ${}^7\mathrm{Be} + e^- \to {}^7\mathrm{Li} + \nu_e$ lines, 862 / 384 keV
- ${}^8\mathrm{B} \to {}^8\mathrm{Be}^* + e^+ + \nu_e$ continuum, ≤ 15 MeV
- ${}^3\mathrm{He} + p \to {}^4\mathrm{He} + e^+ + \nu_e$ *hep*, ≤ 18.8 MeV

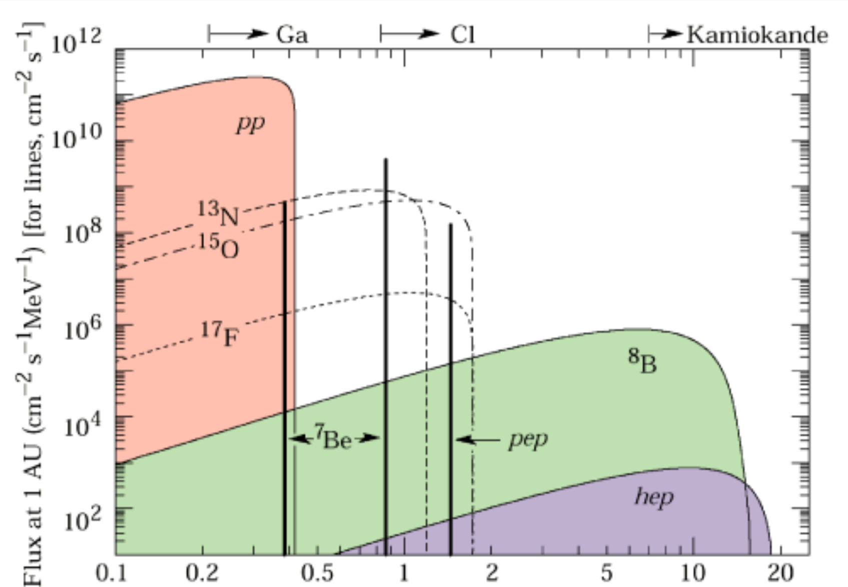
CNO cycle $\sim 1\%$ of $L_\odot$

- ${}^{13}\mathrm{N} \to {}^{13}\mathrm{C} + e^+ + \nu_e$ continuum, ≤ 1199 keV
- ${}^{15}\mathrm{O} \to {}^{15}\mathrm{N} + e^+ + \nu_e$ continuum, ≤ 1732 keV
- ${}^{17}\mathrm{F} \to {}^{17}\mathrm{O} + e^+ + \nu_e$ continuum, ≤ 1740 keV

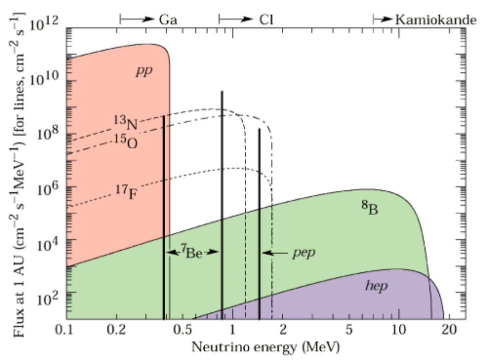
Three-body β^+ decays give *continua*; two-body electron capture gives *lines*. That distinction is the single most useful thing to know when reading a solar neutrino spectrum.

Inference: Each branch carries its own spectral fingerprint. Measure the spectrum, count the neutrinos, and you have measured the branching ratios of hydrogen burning in a working star.

The Solar Neutrino Spectrum



The Solar Neutrino Spectrum — How to Read the Plot

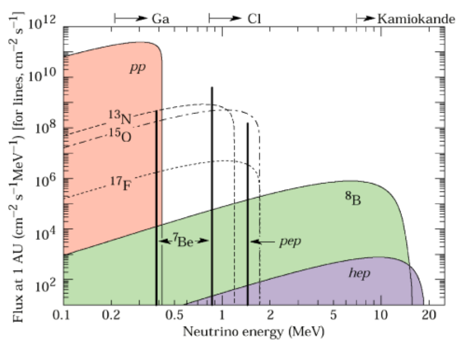


Predicted fluxes at 1 AU.

1. **Both axes are logarithmic.** Energy spans 0.1–20 MeV; flux spans twelve decades. A curve sitting "slightly higher" than another is higher by a factor of a hundred.
2. **Two different units share the vertical axis.** The continua are plotted *per MeV* ($\mathrm{cm}^{-2}\mathrm{s}^{-1}\mathrm{MeV}^{-1}$); the vertical lines are *total* fluxes ($\mathrm{cm}^{-2}\mathrm{s}^{-1}$). So the height of a line is not comparable to the height of a curve. For a continuum the physical flux is the *area* underneath it.
3. **Curve or line is pure kinematics.** A three-body β^+ decay shares the energy $\Rightarrow$ continuum, cut off at the Q-value. A two-body electron capture fixes it exactly $\Rightarrow$ line. Hence $^{7}\mathrm{Be}$ gives two lines, from the ground and first excited states of $^{7}\mathrm{Li}$.
4. **Line style names the cycle.** Solid curves are pp-chain branches; the dashed and dot-dashed curves are the CNO emitters $^{13}\mathrm{N}$, $^{15}\mathrm{O}$ and $^{17}\mathrm{F}$, sitting two to three decades below *pp*.

Check you have read it correctly: the tallest feature on the plot is the *pp* band at low energy, and the $^{8}\mathrm{B}$ band — which extends furthest right — lies roughly four decades below it.

The Same Plot — What It Tells You



1. **The arrows across the top are the history of the field.**
   - Ga, 233 keV — GALLEX and SAGE. The only threshold low enough to reach the primary *pp* neutrinos.
   - Cl, 814 keV — Homestake. Sees $^{7}\mathrm{Be}$ and $^{8}\mathrm{B}$ only, and is blind to the reaction that actually powers the Sun.
   - Kamiokande, $\sim$5 MeV — water Cherenkov. Sees the $^{8}\mathrm{B}$ tail and nothing else.

Everything to the *left* of an arrow is invisible to that experiment.

2. **Where Borexino sits.**
   - Threshold $\sim$0.19 MeV $\Rightarrow$ the whole plot is reachable from one detector — the 2018 pp-chain result.
   - The CNO curves peak below 1 MeV, buried under *pp* and $^{7}\mathrm{Be}$. Detecting them needed background work, not a lower threshold.

Take away

The *pp* flux exceeds $^{8}\mathrm{B}$ by roughly four orders of magnitude, yet $^{8}\mathrm{B}$ was measured decades earlier. Detectability is set by threshold, not abundance.

The Standard Solar Model: What It Is

Every flux quoted in this course is compared against a Standard Solar Model (SSM). It is worth being precise about what that model is and is not.

The construction, in four steps:

1. Solve the four equations of stellar structure (hydrostatic equilibrium, mass conservation, energy generation, energy transport — written out in Part II) for a $1\,M_\odot$ star of assumed initial composition.
2. Supply the input physics: equation of state, radiative opacities, nuclear reaction rates, and element diffusion.
3. Evolve from the zero-age main sequence to $t = 4.57$ Gyr.
4. Calibrate: adjust the initial helium abundance, the initial metal-to-hydrogen ratio, and the mixing-length parameter until the model reproduces the observed present-day $L_\odot$, $R_\odot$ and surface $(Z/X)_\odot$.

The essential point for this course

The neutrino fluxes are not fitted. They are *predictions* that fall out of the calibrated model. That is what makes them a genuine test rather than a consistency check.

Why the Fluxes Are Such Sharp Diagnostics

The different branches respond to the central temperature with wildly different sensitivities. Writing $\Phi \propto T_c^{\beta}$ near the solar value:

Flux	β	What it is most useful for
$\Phi(pp)$	≈ -1.1	essentially fixed by $L_\odot$; a normalisation check
$\Phi(^7\mathrm{Be})$	$\approx +11$	a precise core thermometer
$\Phi(^8\mathrm{B})$	$\approx +24$	the sharpest thermometer available
$\Phi(\mathrm{CNO})$	$\approx +20$	thermometer *and* direct catalyst assay

- $\Phi(pp)$ barely moves because the pp chain must supply the luminosity, and the luminosity is measured. It is anchored.
- $\Phi(^8\mathrm{B}) \propto T_c^{24}$ means a 1% error in T_c becomes a 27% error in the predicted flux. Measuring $^8\mathrm{B}$ to 3% therefore pins T_c to about 0.1%.
- Only the CNO flux carries the additional *linear* dependence on the C and N abundance, which is why it, and nothing else, can weigh the core's catalysts.

The Two Standard Solar Models

Metalicity = $\left(\frac{Z}{X_{\odot}}\right)_{\odot}$ (metal to H ratio)

He & others

$\%_H$

The SSM requires the surface composition as an input, and two independent, high-quality methods disagree about it. Two models therefore coexist.

SSM-HZ (high Z) 1990

Built on older spectroscopic abundances, and supported by helioseismology: millions of p-mode frequencies invert to give the sound-speed profile, the depth of the convective envelope and the surface helium abundance. All three prefer a higher metal content. ⇒ High metals

SSM-LZ (low Z) 2000

Built on modern photospheric spectroscopy: 3D hydrodynamic model atmospheres with non-LTE line formation, which lowered the inferred C, N and O abundances by $\sim 30\%$ in the 2000s. Technically the better atmosphere calculation — but it breaks the seismic agreement. ⇒ low metal

The solar metallicity problem

Metals matter twice. **Directly**, because C, N and O *are* the CNO catalysts. **Indirectly, and more importantly**, because they dominate the radiative opacity, which sets the temperature gradient, which sets T_c, which — through the exponents on the previous slide — moves every flux at once.

This is the question Borexino was ultimately built to settle.

Neutrino Oscillations: From Problem to Tool

Neutrinos are produced in the core as pure ν_e. They do not stay that way. Define the survival probability P_{ee} as the fraction still of electron flavour on arrival.

- **Low energy** ($E_\nu \lesssim 1$ MeV, i.e. *pp*, $^7\mathrm{Be}$): conversion is dominated by vacuum averaging $\Rightarrow P_{ee} \approx 0.55$.
- **High energy** ($E_\nu \gtrsim 5$ MeV, i.e. $^8\mathrm{B}$): passage through the dense electron gas of the solar interior resonantly enhances conversion — the MSW effect $\Rightarrow P_{ee} \approx 0.35$.

Two lessons worth stating explicitly

1. The 1968–1990s “solar neutrino problem” — every experiment seeing too few neutrinos — looked like a failure of nuclear astrophysics. It was in fact the discovery of neutrino mass. Always ask which link in an inference chain is weakest; it is rarely the one under suspicion.

2. Because the oscillation parameters are now measured independently (reactor experiments, KamLAND), flavour conversion has become a known correction rather than an unknown. We can convert a measured ν_e rate back into a production rate in the core. Part II gives the formula.

How Do You “See” a Neutrino?

Given $\sigma \sim 10^{-44}$ cm^2, every solar neutrino experiment ever built has had to satisfy the same three requirements.

1. Go deep

Kilometres of rock overhead. Cosmic rays blind any surface detector instantly.

2. Go big

Tons to kilotons of target — water, D_2O, gallium, chlorine or organic scintillator — to maximise the number of target electrons or nuclei.

3. Go clean

Radiopurity below 10^{-18} g/g in U and Th, so that natural radioactivity does not swamp the signal.

Three detection techniques have been used:

1. **Radiochemical.** A neutrino converts a target atom into a radioactive isotope; the isotope is chemically extracted weeks later and its decays are counted. Gives a *rate above threshold* and nothing else — no energy, no direction, no time.
2. **Water Cherenkov.** A scattered electron exceeding the speed of light in water emits a directional cone of light. Gives energy, direction and time — but only above $\sim$ 3.5–5 MeV.
3. **Liquid scintillation (Borexino).** A scattered electron excites organic molecules, which emit abundant isotropic light. Gives energy, position and particle type down to very low threshold.

1968: Homestake — the Pioneer

1478 m underground in a South Dakota gold mine. The birth of solar neutrino astronomy, and a thirty-year controversy.

Ray Davis Jr. beside the 615-tonne C_2Cl_4 tank.

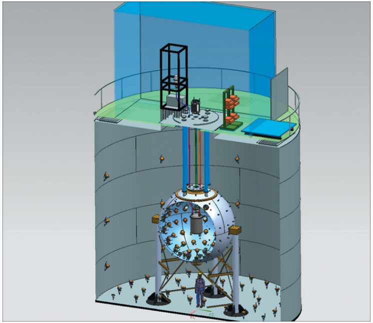
Radiochemical, on chlorine:

$$\nu_e + {}^{37}\mathrm{Cl} \longrightarrow {}^{37}\mathrm{Ar} + e^-$$

- Threshold 814 keV — sensitive to ${}^7\mathrm{Be}$ and ${}^8\mathrm{B}$ only, and completely blind to the primary *pp* neutrinos.
- Every few weeks, helium was purged through the tank to extract $\sim$ 15 radioactive argon atoms from 10^{30} chlorine atoms. It is worth pausing on that ratio.
- Result: about one third of the predicted rate. The **solar neutrino problem** was born, and it was not resolved for three decades.

1990s: GALLEX and SAGE — Reaching the *pp* Neutrinos

Homestake's 814 keV threshold made it blind to the reaction that actually powers the Sun ($E \leq 420$ keV). Gallium was the way past that barrier.



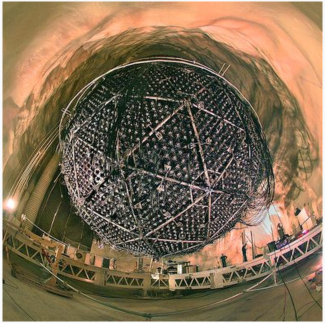
The GALLEX 30-tonne target tank at Gran Sasso.

$$\nu_e + {}^{71}\mathrm{Ga} \longrightarrow {}^{71}\mathrm{Ge} + e^-$$

- Threshold 233 keV — low enough to capture primary *pp* neutrinos for the first time.
- **GALLEX/GNO** (Gran Sasso, Italy): 30 t of gallium as $\mathrm{GaCl_3}$ solution. **SAGE** (Baksan, Russia): 50 t of liquid metallic gallium.
- Detected roughly 60% of the expected rate. The deficit was real, it was energy-dependent, and it was now visible in the dominant branch — a much harder result to blame on stellar modelling.

2001: SNO — the Smoking Gun

2.1 km deep in a Canadian nickel mine; 1000 t of ultra-pure heavy water. The experiment that broke the deadlock, by measuring two things at once.



SNO's 12 m acrylic vessel inside its 9500-PMT sphere.

Heavy water gives three channels:

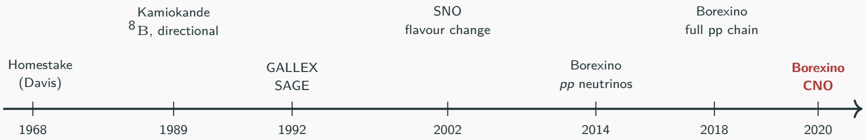
1. **Charged current:** $\nu_e + d \to p + p + e^-$
   $\Rightarrow$ sensitive to ν_e only.
2. **Neutral current:** $\nu_x + d \to p + n + \nu_x$
   $\Rightarrow$ sensitive equally to all flavours.
3. **Elastic scattering:** $\nu_x + e^- \to \nu_x + e^-$.

The decisive comparison

$\Phi_{\mathrm{CC}}(\nu_e) < \Phi_{\mathrm{NC}}(\nu_{\mathrm{total}})$, while Φ_{NC} matched the Standard Solar Model.

The Sun was right all along. The missing neutrinos had changed flavour. Nobel Prize, 2015.

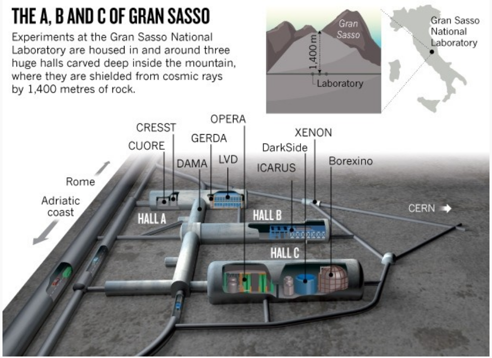
Sixty Years in One Line



- **1968–1990s.** Homestake, Kamiokande, GALLEX and SAGE all see fewer neutrinos than predicted. Is the Sun wrong, or is the neutrino wrong?
- **2001–2002.** SNO separates charged- and neutral-current rates. The *total* flux matches the solar model; the *electron-flavour* flux does not. Neutrinos oscillate.
- **2014–2018.** Borexino performs complete spectroscopy of the pp chain — *pp*, *pep*, ^{7}Be, ^{8}B — confirming that $\sim$ 99% of solar power comes from the pp chain. $\rightarrow$ Section I.2
- **2020–2023.** The last missing piece: CNO. $\rightarrow$ Section I.3

Deep inside the Apennines in Abruzzo, Italy, the **Laboratori Nazionali del Gran Sasso** (LNGS) is the largest underground laboratory in the world, and the site of a disproportionate share of solar fusion history.

The Gran Sasso massif housing the underground halls.



Experimental halls at Gran Sasso.

- **Natural shielding:** 1400 m of rock overhead (3800 m water equivalent) suppresses the cosmic muon flux by a factor of 10^6.
- **Home to three complementary milestones:**
  - **GALLEX/GNO** — first detection of primary *pp* neutrinos.
  - **LUNA** — an accelerator underground, measuring nuclear cross sections *at true solar energies* rather than extrapolating to them. This is the direct experimental input to Part II.
  - **Borexino** — complete pp-chain spectroscopy and the first CNO detection.

Note the pairing: the same mountain hosts the experiment that measures the reaction rates and the experiment that measures their consequences.

Why Borexino Stands Out

SNO settled oscillations. So why was another detector needed?

Water Cherenkov (SNO, Super-K)

- Threshold $E \gtrsim 3.5$–5 MeV.
- Sees only the top $\sim 0.01\%$ of the solar neutrino spectrum — the $^8\mathrm{B}$ branch.
- Structurally blind to *pp*, *pep*, $^7\mathrm{Be}$ and CNO: the Cherenkov light yield is too faint for natural radioactivity to be distinguished from signal at low energy.

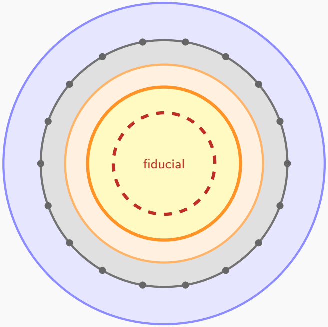
Liquid scintillator (Borexino)

- Emits $\sim 50\times$ more light per MeV: about 500 photoelectrons per MeV.
- Threshold pushed down to ~ 0.19 MeV.
- Can therefore perform spectroscopy across the entire solar neutrino spectrum — *pp*, *pep*, $^7\mathrm{Be}$, $^8\mathrm{B}$ and CNO — in a single detector.

The trade, stated plainly

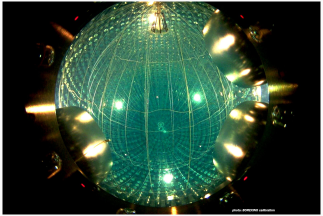
Scintillation light is isotropic, so Borexino gives up the directional information that Cherenkov detectors get for free. In exchange it gains a factor of 50 in light yield and a factor of 20 in threshold. Everything Borexino achieved, and every difficulty it faced, follows from that single trade.

Borexino: Architecture



"Onion-skin" shielding, inside out:

- Inner nylon vessel, $R = 4.25$ m: 280 t of pseudocumene + PPO scintillator.
- Second nylon vessel, $R = 5.50$ m: buffer liquid, blocking inward radon diffusion.
- Stainless steel sphere, $R = 6.85$ m: carries 2212 PMTs.
- Water tank: passive shield plus 208 PMTs acting as an active Cherenkov muon veto.



Inside the stainless steel sphere.

The design principle

Each layer exists to protect the one inside it, and the innermost ~ 71 t alone is used for analysis. Roughly three quarters of the scintillator is sacrificed as shielding for the remaining quarter. Purity is bought with volume.

From a Photon to a Number

Borexino detects the electron a neutrino happens to hit:

$$\nu_{e,\mu,\tau} + e^- \longrightarrow \nu_{e,\mu,\tau} + e^- \qquad \text{(elastic scattering)}$$

The recoiling electron ionises the scintillator; the emitted light is collected. Three independent observables are extracted from that light.

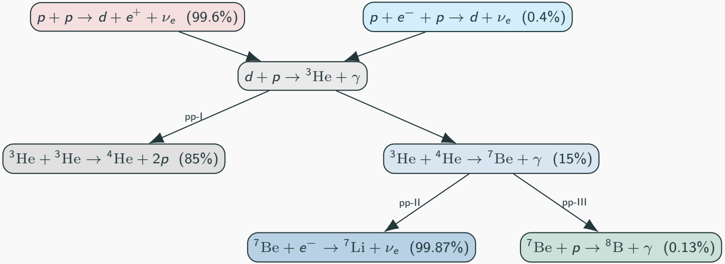
- **Energy** $\leftarrow$ total photon count (N_h, normalised to 2000 live PMTs). Resolution $\sigma_E/\sqrt{E} \approx 6\%$ at 1 MeV.
- **Position** $\leftarrow$ photon arrival times, triangulated by time-of-flight. $\sigma \approx 11$ cm at 1 MeV. This is what makes a fiducial cut possible, and it separates uniform internal events from external γ-rays that cluster near the vessel.
- **Particle type** $\leftarrow$ the time profile of the pulse. α, β^- and β^+ quench differently and produce measurably different pulse shapes. Hold on to this: both Borexino results depend on it.

What we actually measure

The raw output is a histogram of counts versus deposited energy, nothing more. Individual components are separated only by fitting their known theoretical spectral shapes simultaneously. There is no event-by-event tag saying “this one was a neutrino”.

I.2 The pp Chain — Borexino 2018

The Nuclear Roadmap of the pp Chain



- ***pp*:** continuous, $\leq$ 420 keV. About 90% of all solar neutrinos.
- ***pep*:** monoenergetic line at 1.44 MeV. Same initial state as *pp* but with an electron in the entrance channel — a direct competitor to the primary reaction, and rigidly tied to it.
- **${}^7\mathrm{Be}$:** two lines, 862 keV (90%) and 384 keV (10%), depending on whether ${}^7\mathrm{Li}$ is left in its ground or excited state.
- **${}^8\mathrm{B}$:** broad, $\leq$ 15 MeV. A rare branch, but historically the easiest to see.

Dividing the Spectrum: LER and HER

Backgrounds vary by orders of magnitude across the spectrum, so a single analysis cannot work. Borexino splits the problem in two.

Low-Energy Region (LER)

- Window: 0.19 – 2.93 MeV.
- Targets: *pp*, $^{7}\mathrm{Be}$, *pep*.
- Exposure: 1291.51 d $\times$ 71.3 t (Phase-II).
- Fiducial cut: $R < 2.8$ m, $-1.8 < z < 2.2$ m, to reject external γ from $^{40}\mathrm{K}$, $^{214}\mathrm{Bi}$, $^{208}\mathrm{Tl}$.

High-Energy Region (HER)

- Window: 3.2 – 16.0 MeV.
- Target: $^{8}\mathrm{B}$.
- Exposure: 2062.4 d $\times$ 227.8 t (Phase-I + II).
- Subdivided: HER-I (3.2–5.7 MeV, $z < 2.5$ m, 227.8 t); HER-II (5.7–16 MeV, full 266.0 t).

Why 3.2 MeV, and not lower?

The threshold is set by a specific line: the 2.614 MeV γ from $^{208}\mathrm{Tl}$ in the vessel nylon. Everything below it is unusable for $^{8}\mathrm{B}$. Note how much larger a fiducial volume becomes usable once you are above the natural radioactivity — 266 t at high energy against 71 t at low energy. Purity requirements are energy-dependent.

Multivariate Fitting in the Low-Energy Region

Electron recoil spectra have no sharp peaks — only Compton-like edges. So *pp*, $^{7}\mathrm{Be}$, *pep* and every background must be separated *simultaneously*, using three independent handles at once.

- **Energy spectrum.** Fit N_h for both the $^{11}\mathrm{C}$-tagged and $^{11}\mathrm{C}$-subtracted datasets (threefold coincidence, TFC).
- **Spatial distribution.** Internal events are uniform in the fiducial volume; external γ-rays rise steeply with radius. The radial profile separates them.
- **Pulse-shape discrimination.** Separates electron-like recoils (ν, β^-) from positron-like events ($^{11}\mathrm{C} \to e^+$), because positron annihilation leaves a distinct time profile.

Three orthogonal discriminants on the same events is what makes an otherwise degenerate fit tractable.

Internal backgrounds fitted

$^{14}\mathrm{C}$	40 ± 2 Bq/100 t
$^{85}\mathrm{Kr}$	6.8 ± 1.8 cpd/100 t
$^{210}\mathrm{Bi}$	17.5 ± 1.9 cpd/100 t
$^{11}\mathrm{C}$	26.8 ± 0.2 cpd/100 t
$^{210}\mathrm{Po}$	260 ± 3 cpd/100 t

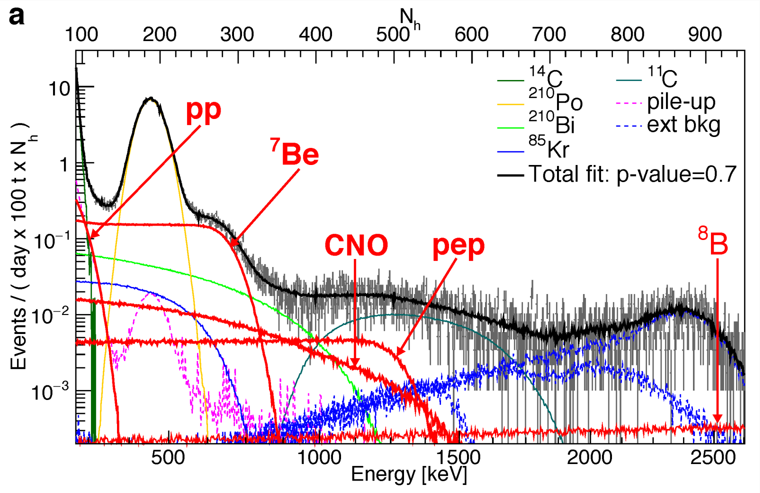
Note $^{14}\mathrm{C}$: an *irreducible* contaminant of any organic scintillator, and the reason the threshold cannot go below ~ 0.19 MeV.

Measured Rates and Fluxes

Solar ν	Measured rate (cpd / 100 t)	Measured flux ($cm^{-2}s^{-1}$)	SSM-HZ	SSM-LZ
pp	$134 \pm 10\,^{+6}_{-10}$	$(6.1 \pm 0.5\,^{+0.3}_{-0.5}) \times 10^{10}$	5.98×10^{10}	6.03×10^{10}
$^{7}\mathrm{Be}$	$48.3 \pm 1.1\,^{+0.4}_{-0.7}$	$(4.99 \pm 0.11\,^{+0.06}_{-0.08}) \times 10^{9}$	4.93×10^{9}	4.50×10^{9}
pep (HZ)	$2.43 \pm 0.36\,^{+0.15}_{-0.22}$	$(1.27 \pm 0.19\,^{+0.08}_{-0.12}) \times 10^{8}$	1.44×10^{8}	1.46×10^{8}
pep (LZ)	$2.65 \pm 0.36\,^{+0.15}_{-0.24}$	$(1.39 \pm 0.19\,^{+0.08}_{-0.13}) \times 10^{8}$	1.44×10^{8}	1.46×10^{8}
$^{8}\mathrm{B}$	$0.223\,^{+0.015}_{-0.016}\,^{+0.006}_{-0.006}$	$(5.68\,^{+0.39}_{-0.41}\,^{+0.03}_{-0.03}) \times 10^{6}$	5.46×10^{6}	4.50×10^{6}
hep	< 0.002 (90% CL)	$< 2.2 \times 10^{5}$ (90% CL)	7.98×10^{3}	8.25×10^{3}
CNO	< 8.1 (95% CL)	$< 7.9 \times 10^{8}$ (95% CL)	4.88×10^{8}	3.51×10^{8}

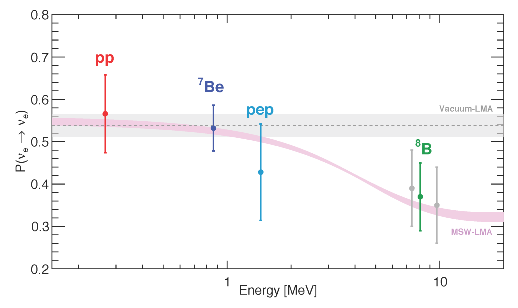
- $^{7}\mathrm{Be}$ **precision: 2.7%** — about twice as precise as the solar models being tested. When the measurement beats the theory, the measurement starts constraining the theory.
- *pep* **detected:** the no-*pep* hypothesis is rejected at $> 5\sigma$.
- **CNO in 2018:** only an upper limit, < 8.1 cpd/100 t. That limit sits *above* both model predictions and so decides nothing — which is precisely the gap the 2020 measurement was built to close.

Measured Rates and Fluxes



Distribution of events after selection cuts and the fit of the Borexino Low-Energy Region (LER, [0.19–2.93] MeV) spectrum. The individual solar neutrino components (*pp*, ^{7}Be, *pep*, and CNO) and background sources are depicted over the Three-Fold Coincidence (TFC) subtracted energy spectrum, where the cosmogenic ^{11}C background is suppressed.

Testing Oscillations Across Two Regimes at Once



Electron neutrino survival probability P_{ee} as a function of neutrino energy, adapted from Fig. 3 of *Nature* (2018)

Grey band: pure vacuum oscillation, which predicts a *flat* P_{ee}. Red band: MSW-LMA, which predicts a downward step between 1 and 5 MeV.

- Vacuum-dominated regime, $E_\nu < 1$ MeV: $P_{ee} \approx 0.55$.
- Matter-dominated regime, $E_\nu > 5$ MeV: $P_{ee} \approx 0.35$.
- Pure vacuum-LMA is disfavoured at 98.2% CL (2.4σ) — from a *single detector* spanning both regimes, with correlated systematics that largely cancel.

Two Solar Model Tests That Need No New Physics

1. Is the Sun in thermal equilibrium?

Using only Borexino's own pp-chain fluxes, the nuclear power output is

$$L_{\mathrm{nuc}} = \left(3.89\,^{+0.35}_{-0.42}\right) \times 10^{33}\ \mathrm{erg\ s}^{-1} \qquad \text{versus} \qquad L_{\mathrm{phot}} = 3.846 \times 10^{33}\ \mathrm{erg\ s}^{-1}$$

The two agree. Since photons take $\sim 10^5$ yr to emerge but neutrinos take 8 minutes, this compares the Sun's power output now with its output 10^5 years ago. The Sun has been in thermodynamic equilibrium over that interval — a statement no other measurement can make.

2. Where does the chain terminate?

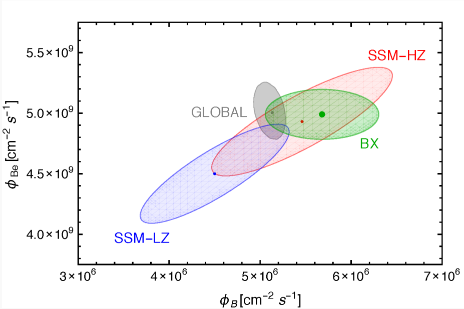
The branching between ${}^3\mathrm{He} + {}^3\mathrm{He}$ (pp-I) and ${}^3\mathrm{He} + {}^4\mathrm{He}$ (pp-II/III) is

$$R_{\mathrm{I/II}} = \frac{2\,\Phi({}^7\mathrm{Be})}{\Phi(pp) - \Phi({}^7\mathrm{Be})} = 0.178\,^{+0.027}_{-0.023}$$

against predictions of 0.180 ± 0.011 (HZ) and 0.161 ± 0.010 (LZ). A branching ratio of stellar nucleosynthesis, measured in a laboratory.

The Metallicity Hint from the pp Chain

$\Phi(^7\mathrm{Be})$ and $\Phi(^8\mathrm{B})$ both depend steeply on T_c, which depends on opacity, which depends on metallicity. Plotting one against the other separates the two models.



Comparison of the Borexino experimental results with the theoretical predictions of the Standard Solar Models for high metallicity (SSM-HZ, GS98) and low metallicity (SSM-LZ, AGSS09) in the $(R_{^7\mathrm{Be}}, R_{^8\mathrm{B}})$ rate space. The elliptical contours represent the 68%, 95%, and 99.7% C.L. experimental allowed regions alongside the theoretical 1σ uncertainty contours for both solar metallicity hypotheses.

- A frequentist likelihood-ratio test disfavours low metallicity at 96.6% CL (2.1σ).
- A Bayesian analysis gives a Bayes factor $B = 4.9$ in favour of high metallicity — "substantial" on the usual scale, but well short of decisive.
- **Caveat, and it matters:** combining Borexino's pp-chain data with global solar and KamLAND data *weakens* the hint, because of tensions in externally determined oscillation parameters.

Inference: The pp chain can only reach metallicity *indirectly*, through T_c. To measure the catalysts themselves you need the one flux that depends on them linearly — CNO.

Summary of Section I.2

1. Borexino Phase-II achieved the first complete spectroscopy of the solar pp chain — *pp*, *pep*, $^{7}\mathrm{Be}$, $^{8}\mathrm{B}$ — inside a single detector.
2. The *pp* flux was measured down to 0.19 MeV; $^{7}\mathrm{Be}$ reached 2.7% precision, better than the solar models it tests.
3. P_{ee} was mapped across both the vacuum- and matter-dominated regimes, confirming MSW-LMA and rejecting pure vacuum oscillation at 98.2% CL.
4. Neutrino-inferred nuclear luminosity matches the photon luminosity, showing the Sun has been in thermal equilibrium over 10^5 years; the pp-I/pp-II branching ratio came out at 0.178.
5. An early 2.1σ hint favoured high-metallicity models — and left CNO as the only remaining way to settle the question.

I.3 The CNO Cycle — Borexino 2020 and 2023

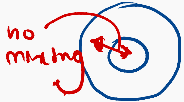
Why CNO Was Worth Four More Years of Work

Section I.2 ended with an upper limit. Three arguments made it worth pushing on to a measurement.

1. **It is the dominant hydrogen-burning mode in the Universe.** The Sun is atypical. Above $M \gtrsim 1.3\,M_{\odot}$ the core is hot enough that CNO takes over, and because massive stars are far more luminous, most hydrogen fused into helium *anywhere* is fused by CNO. Yet the cycle, proposed by Bethe and von Weizsäcker in 1937–39, had never been directly observed.
2. **It is the only flux that weighs the catalysts directly.** Every other flux reaches metallicity indirectly through T_c. Only CNO carries a linear dependence: for CNO flux depends on C & N in core

$$\Phi_{\mathrm{CNO}} \;\propto\; N_{\mathrm{C+N}}(\mathrm{core}) \times f(T_c)$$

3. **The core composition is otherwise unreachable.** A radiative zone separates core from surface with no mixing, so the core still records the initial composition of the protosolar cloud, 4.6 Gyr ago. There is no other way to measure it.

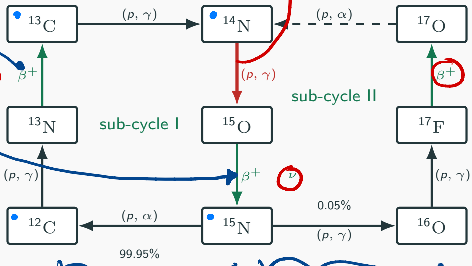


The CNO Bi-cycle and Its Three Neutrinos

$4p \longrightarrow {}^4He + 2e^+ + 2\nu_e + \gamma$

Lowest x section/ S-factor in the cycle 1

β+ decay



β+ decay

99.95% 0.05%

Four protons in, one ^{4}He out, ^{12}C returned unchanged — a true catalytic cycle. Three β^+ decays give three neutrino sources: ^{13}N, ^{15}O, ^{17}F. Sub-cycle I dominates overwhelmingly in the Sun. The red arrow, $^{14}N(p,\gamma)^{15}O$, is the slowest step and therefore sets the pace of the whole cycle; Part II shows why, and what it does to the equilibrium abundances.

each β^+ decay → continuous e^ν neutrino spectrum

$^{13}N \rightarrow E_\nu < 1199$ keV
$^{15}O \rightarrow E_\nu < 1732$ keV
$^{17}F \rightarrow E_\nu < 1740$ keV

The Size of the Target

The two models differ, and the entire scientific question lives in the gap between them.

Model	Predicted CNO rate in Borexino
SSM-LZ (low metallicity)	3.52 ± 0.52 cpd/100 t
SSM-HZ (high metallicity)	4.92 ± 0.78 cpd/100 t
Separation	≈ 1.4 cpd/100 t

100 tons in liquid scintillator

This 1 event/day creates solar metallicity problem → statistical problem / bkg

The models differ by about one and a half events per day in a hundred tonnes of liquid — roughly one nuclear event per day per 70 tonnes.

That single number explains why the 2020 paper spends most of its length discussing thermal insulation rather than the Sun.

Putting the Rates Side by Side

About 6×10^{10} solar neutrinos cross every square centimetre of the detector per second. The tiny cross section turns that into a handful of interactions per day.

major background radiostopes

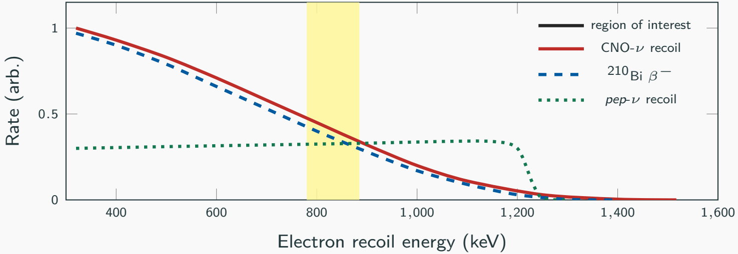
	Rate (cpd/100 t)
CNO neutrino signal (sought)	**a few**
pep neutrinos	2.74 ± 0.04
$^{210}\mathrm{Bi}$ contamination	up to ~ 11.5
$^{11}\mathrm{C}$ (cosmogenic)	tens
$^{85}\mathrm{Kr}$, $^{40}\mathrm{K}$, external γ	few to tens
Ordinary untreated material	$\sim 10^4$–10^5 *per tonne per second*

Inference: The signal is not merely small compared to the background. It is smaller than the *uncertainty* on the background. Everything in the rest of this section is an attempt to fix that.

The Degeneracy That Nearly Killed the Measurement

Three components produce nearly identical electron recoil spectra over the same energy range. A simultaneous fit can trade them against one another almost freely.

Background
→ e^{n} recoils
↳ β^- decay from Bi
↳ pep



Schematic. The CNO and $^{210}\mathrm{Bi}$ shapes are so alike that the spectral fit alone cannot separate them. Two of these three must be pinned down independently, or the third cannot be measured at all.

Two Backgrounds Down, One to Go

1. ***pep* neutrinos — constrained by physics.** The *pep* reaction is an alternative first step of the pp chain, and its rate is rigidly tied to the ordinary *pp* rate. Using the solar luminosity constraint, the known *pp*/*pep* ratio and measured oscillation parameters, the rate is fixed to 2.74 ± 0.04 cpd/100 t — 1.4% precision.

Note the elegance: $L_\odot$ depends only very weakly on CNO, so this constraint does *not* secretly assume the answer. This is worth labouring, because students reliably suspect circularity here.

why ?

2. $^{11}\mathrm{C}$ **— vetoed by coincidence.** Cosmic muons that survive the rock spall carbon in the scintillator:

$$\mu + {}^{12}\mathrm{C} \to {}^{11}\mathrm{C} + n, \qquad n + p \to d + \gamma$$

$^{11}\mathrm{C}$ then β^+ decays with $\tau = 30$ min, squarely in the signal region. The threefold coincidence tags the muon, the neutron-capture γ and the later decay — three events linked in space and time — and removes them.

3. $^{210}\mathrm{Bi}$ **— no such luck.** It is inside the liquid, it is continuously replenished, and it looks like the signal. This one needed a genuinely new idea.

Why $^{210}\mathrm{Bi}$ Cannot Be Measured Directly

It sits in the middle of a decay chain fed by long-lived $^{210}\mathrm{Pb}$:

$$^{210}\mathrm{Pb} \xrightarrow{\beta^-,\ 22.3\ \text{yr}} {}^{210}\mathrm{Bi} \xrightarrow{\beta^-,\ 5.0\ \text{d}} {}^{210}\mathrm{Po} \xrightarrow{\alpha,\ 138.4\ \text{d}} {}^{206}\mathrm{Pb}$$

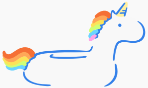
$^{210}\mathrm{Bi}$ — invisible: looks like signal

$^{210}\mathrm{Po}$ — visible: α, taggable

invisible means end point is not detectable

- The $^{210}\mathrm{Pb}$ β endpoint is 63.5 keV — far below the 320 keV analysis threshold. Invisible.
- $^{210}\mathrm{Bi}$ β^- electrons are indistinguishable from neutrino recoils. Invisible *as bismuth*.
- $^{210}\mathrm{Po}$ emits an α. Alphas ionise densely, quench more, and give a distinct pulse shape — so they *can* be tagged event by event.

Inference: If the chain is in secular equilibrium, the $^{210}\mathrm{Po}$ α rate equals the $^{210}\mathrm{Bi}$ rate. Count the alphas you can see to measure the betas you cannot.



Why the Trick Does Not Work Straight Away

Secular equilibrium requires that the polonium you count is the polonium your bismuth produced. In practice it is not.

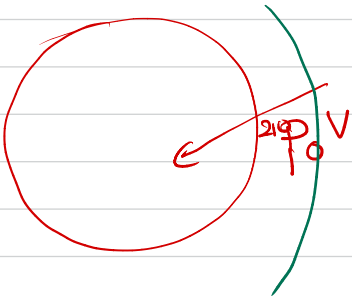
- There are two polonium populations: ${}^{210}\mathrm{Po}^{S}$, supported by ${}^{210}\mathrm{Pb}$ dissolved in the liquid (this one *is* in equilibrium with the bismuth), and ${}^{210}\mathrm{Po}^{V}$, detaching from ${}^{210}\mathrm{Pb}$ deposited on the inner surface of the nylon vessel (pure contamination).
- Diffusion alone would be harmless: over one half-life, polonium diffuses far less than the ≈ 1 m from vessel to fiducial volume.
- **Convection is not harmless.** Temperature gradients drive slow currents that physically carry vessel polonium inward.
- ${}^{210}\mathrm{Bi}$, with its 5-day half-life, decays before it can be transported — so the contamination corrupts the tracer but not the thing being traced.

Hence an upper limit rather than an equality:

$$R({}^{210}\mathrm{Po}_{\mathrm{min}}) = R({}^{210}\mathrm{Bi}) + R({}^{210}\mathrm{Po}^{V}), \qquad R({}^{210}\mathrm{Po}^{V}) \geq 0$$

→ when fluid has diff temp. in diff parts in tank → density difference cause light fluid to rise cooler fluid m sink

→ thermal convection



~ vessal polonium

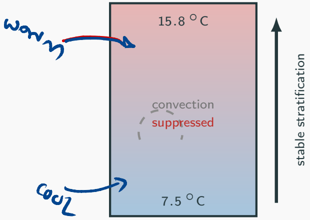
dragged in to central vessal fluid volume

The Fix Is Thermal Engineering, Not Physics

to map ^{210}Po bottleneck

Suppressing convection requires a stable vertical temperature gradient: cold at the bottom, warm at the top. Between 2015 and 2019 the collaboration essentially rebuilt the detector's thermal environment.

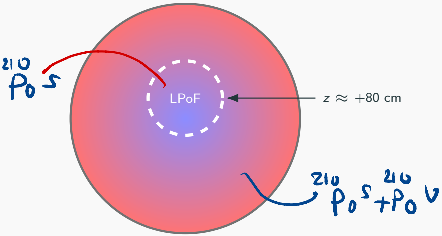
- **2015:** 900 m^2 of mineral-wool insulation wrapped around the water tank; water recirculation switched off to let the fluid stratify.
- **2016:** active temperature control system on the dome — 3 kW heater, circulation pump, PID controller.
- **2019:** the *entire experimental hall* air supply brought under control; a 70 kW heater regulating inlet air to within 0.05 °C.
- 66 probes at 0.07 °C resolution monitor the structure continuously.



Established gradient: 7.5 °C at the floor, 15.8 °C at the top, so $\Delta T/\Delta z \approx 0.5\,°\mathrm{C\ m^{-1}}$.

The Low Polonium Field (LPoF)

Once convection was quelled, the polonium distribution settled into a stable pattern with a clear minimum slightly above the detector's equator, at $z \approx +80$ cm. This is the Low Polonium Field (LPoF): where vessel contamination has least reach, and the measured polonium is closest to pure, bismuth-supported polonium.



Schematic: blue = low $^{210}\mathrm{Po}$, red = high.

The minimum was located by fitting the 3D polonium activity with a paraboloid, cross-checked with cubic splines and with solutions of the diffusion equation.

Evidence it is a real physical structure, not a statistical fluke:

- It moves by less than 20 cm per month — a quasi-steady fluid state.
- Its shape and position are reproduced by independent fluid-dynamics simulations.

Result:

$$R(^{210}\mathrm{Po}_{\mathrm{min}}) = 11.5 \pm 1.0 \text{ cpd}/100\,\mathrm{t}$$

Extrapolated over the fiducial volume, with uniformity systematics in quadrature:

$$\boxed{R(^{210}\mathrm{Bi}) \leq 11.5 \pm 1.3 \text{ cpd}/100\,\mathrm{t}}$$

A Cross-Check Worth Teaching

Extrapolating the LPoF result to the whole fiducial volume is only legitimate if $^{210}\mathrm{Bi}$ is uniform in space and stable in time. Several checks were performed; one is self-contained enough to show a class.

Earth's orbit as a calibration source

Earth's orbit is slightly eccentric, so the solar neutrino flux at Earth varies by 3.3% over the year — a purely geometric, exactly known modulation.

Borexino sees that 3.3% sinusoid in its β-like event rate, dominated by $^{7}\mathrm{Be}$ neutrinos.

Inference: If a 3.3% annual variation is visible above the noise, the backgrounds underneath it must be stable at better than that level. The Earth's orbit has been used to certify the detector's own stability — and, incidentally, to confirm once more that the signal really comes from the Sun.

How the Number Was Extracted

Dataset: Borexino Phase-III, July 2016 to February 2020 — 1072 days of live time in a 71.3 t fiducial volume, analysed between 320 and 2640 keV.

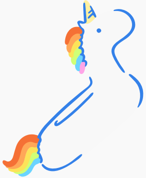
Method: a multivariate fit performed simultaneously on the *energy* spectrum and the *radial* distribution, with the data split into $^{11}\mathrm{C}$-subtracted and $^{11}\mathrm{C}$-enriched samples via the threefold coincidence tag. Reference spectral shapes come from a full Geant4 simulation of the detector.

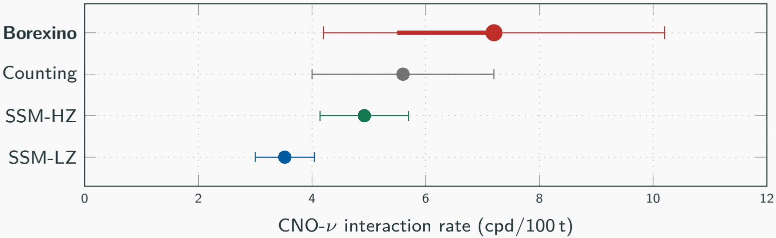
The two constraints that make it possible:

- *pep* rate fixed at 2.74 ± 0.04 cpd/100 t with a symmetric Gaussian penalty.
- $^{210}\mathrm{Bi}$ left free between 0 and 11.5 cpd/100 t, above which a half-Gaussian penalty applies. This asymmetry is the reason the final likelihood profile is lopsided — see two slides on.

Fit quality: *P*-value = 0.3. Data and model agree.

The Result

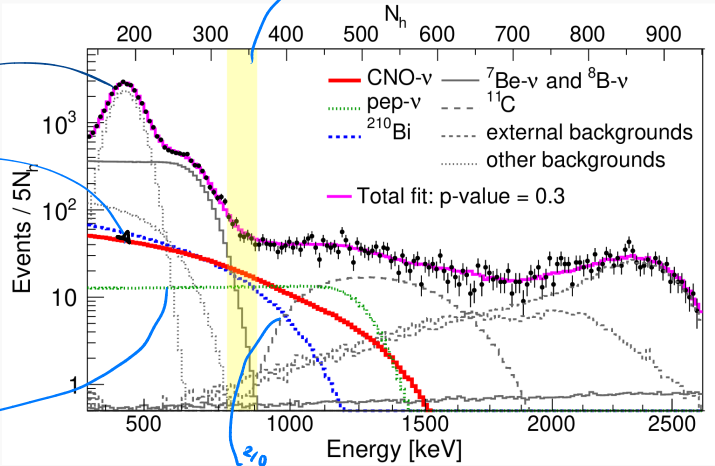




Interaction rate	$7.2\,^{+3.0}_{-1.7}$ cpd/100 t
Converted flux at Earth	$7.0\,^{+3.0}_{-2.0} \times 10^{8}$ cm^{-2}s^{-1}
Significance (profile likelihood)	5.1σ
Independent counting analysis	5.6 ± 1.6 cpd/100 t, i.e. 3.5σ
Compatibility	HZ at 0.5σ, LZ at 1.3σ
No-CNO hypothesis	**excluded at $> 5.0\sigma$ (99% CL)**

Phase III

this region makes CNO contributes mostly before dropoff



Full fit

CNO neutrino recoil component

11C component

210Bi component

Reading the Result Honestly

This is where students should be taught care: the headline and the fine print say different things.

What was firmly established

The CNO cycle runs in the Sun. The zero-CNO hypothesis is dead at more than 5σ. This is the first direct experimental evidence for the mechanism Bethe and von Weizsäcker proposed in the 1930s — eighty years between prediction and detection.

What was not settled in 2020

The metallicity problem stayed open. The measured rate is compatible with *both* models: 0.5σ from HZ, 1.3σ from LZ. The error bar was simply too wide to referee between them. Only combined with Borexino's other fluxes did LZ become disfavoured, and then at just 2.1σ.

Steep on the low side: the half-Gaussian ceiling on $^{210}\mathrm{Bi}$ means the fit cannot invent extra bismuth to absorb the counts.

Shallow on the high side: CNO and $^{210}\mathrm{Bi}$ remain nearly degenerate, so the data cannot distinguish a high rate from a moderate one.

Systematics: 2500 refits with varied ranges and binnings; $> 10^6$ pseudo-datasets with deformed energy scale; three alternative $^{210}\mathrm{Bi}$ β shapes differing by up to 18%. Total $^{+0.6}_{-0.5}$ cpd/100 t — the measurement is statistics-limited, not systematics-limited.

What Happened Next: Directionality, 2022–2023

Borexino found an entirely independent handle, one that removes the bismuth problem rather than bounding it: Correlated and Integrated Directionality (CID).

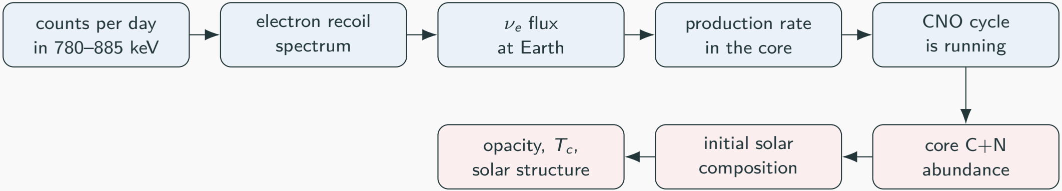
- In a liquid scintillator, abundant isotropic scintillation light swamps the sparse directional Cherenkov light — but the first few photons to arrive still remember the direction.
- Summed over many events, this correlates the signal with the Sun's position in the sky.
- Crucially, it separates solar neutrinos from radioactivity with no assumption about $^{210}\mathrm{Bi}$ at all, and can use the entire 2007–2021 dataset rather than only Phase-III.

Analysis	Rate (cpd/100 t)	Metallicity verdict
Nature **587**, 577 (2020), spectral	$7.2^{+3.0}_{-1.7}$	LZ disfavoured at 2.1σ (combined)
Phys. Rev. D **108**, 102005 (2023), CID	$6.7^{+1.2}_{-0.8}$	LZ disfavoured at 3.2σ

Core abundance from the final result: $N_{\mathrm{CN}} = 5.81^{+1.22}_{-0.94} \times 10^{-4}$ relative to hydrogen.

The Sun's core now appears metal-rich, siding with helioseismology — but not yet at a level a physicist would call decisive.

The Inference Chain, Written Out



Where model dependence enters — be explicit with students:

(a)→(b) essentially pure measurement.

(b)→(c) needs the ν–e elastic scattering cross section. Electroweak theory; known to well under a percent. *Part II.6*

(c)→(d) needs the MSW oscillation parameters. Measured elsewhere, treated as a known correction. *Part II.6*

(d)→(f) needs the CNO reaction rates and the core temperature. This is the weakest link, and it is exactly what Part II is for. *Parts II.1–II.5*

2018 versus 2020: Two Different Experiments

	2018 — pp chain	**2020 — CNO cycle**
Primary target	*pp*, *pep*, $^{7}\mathrm{Be}$, $^{8}\mathrm{B}$ fluxes	CNO flux ($^{13}\mathrm{N}$, $^{15}\mathrm{O}$, $^{17}\mathrm{F}$)
Share of $L_{\odot}$	$\sim$ 99%	$\sim$ 1%
Data period	Phase-II (2011–2016)	Phase-III (2016–2020)
Core strategy	multivariate fit, LER + HER	thermal stabilisation + LPoF
Main obstacle	$^{14}\mathrm{C}$, $^{11}\mathrm{C}$, $^{210}\mathrm{Bi}$	internal $^{210}\mathrm{Bi}$ and convection
Chief physics impact	test the MSW effect	measure the core metallicity
Limiting factor	systematics and purity	counting statistics

The methodological lesson

The 2018 result came from analysing the data better. The 2020 result came from changing the detector's environment so that a background became bounded. When a measurement is degenerate, no amount of statistical sophistication will save it — you have to break the degeneracy physically. The 2023 CID analysis makes the same point a third time, by finding an entirely new observable in data that had already been collected.

Summary of Part I

1. Stellar interiors are hidden behind 10^5 yr of photon diffusion. Neutrinos are the only direct probe of stellar nuclear reactions.
2. Every branch of hydrogen burning emits a neutrino with a characteristic spectral signature, so neutrino spectroscopy is branching-ratio spectroscopy.
3. Sixty years of detectors — Homestake, GALLEX/SAGE, Kamiokande, SNO — first produced the solar neutrino problem and then resolved it as neutrino oscillation, converting flavour conversion from a mystery into a calibrated tool.
4. Borexino's liquid scintillator traded directionality for a factor of 50 in light yield, which alone made low-energy spectroscopy possible. In 2018 it measured the complete pp chain, confirmed MSW-LMA, and showed $L_{\mathrm{nuc}} = L_{\mathrm{phot}}$.
5. In 2020 it detected CNO neutrinos at 5.1σ, $7.2^{+3.0}_{-1.7}$ cpd/100 t — made possible not by a new detector but by thermally stratifying the existing one to freeze convection.
6. Because the CNO rate scales with core C and N abundance, this is the first direct probe of the Sun's primordial core metallicity. With the 2023 directional analysis it leans towards the high-metallicity models favoured by helioseismology.

Problems and Discussion — Part I

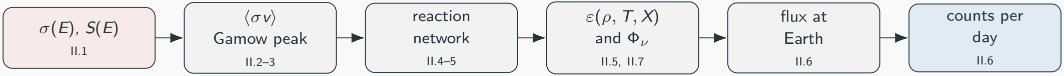
1. The Sun's luminosity is 3.85×10^{26} W and each $4p \to {}^4\mathrm{He}$ releases 26.7 MeV together with two neutrinos. Estimate the total solar neutrino flux at Earth and compare with the quoted 6×10^{10} $\mathrm{cm}^{-2}\mathrm{s}^{-1}$. Why is your answer close *without* any nuclear physics input?
2. The *pp* neutrino flux is roughly 100 times the CNO flux, yet CNO was detected six years later. Why is flux the wrong figure of merit?
3. Why is the ${}^8\mathrm{B}$ threshold set at 3.2 MeV when *pp* and ${}^7\mathrm{Be}$ are measured down to 0.19 MeV? Name the specific background.
4. Explain carefully why using the solar luminosity to constrain the *pep* rate is *not* circular when the goal is to measure CNO.
5. Explain why $P_{ee} \approx 0.57$ at low energy and ≈ 0.37 at high energy demonstrates matter-enhanced conversion rather than a normalisation error.
6. Suppose someone reports a CNO rate of zero. Which single measurement contradicts them immediately, and why?
7. If the detector had never been thermally insulated, what would the limit on ${}^{210}\mathrm{Bi}$ have been, and what would that have done to the significance?
8. The measurement is statistics-limited. Roughly how much more exposure would separate HZ from LZ at 3σ? Was building a new analysis (CID) the better move?

PART II

The Theory Needed to Read Those Observations

What Part II Has to Deliver

Part I quoted numbers: a rate of 7.2 cpd/100 t, a flux of 7.0×10^8 cm^{-2}s^{-1}, a branching ratio of 0.178, a temperature sensitivity T_c^{24}. None of those means anything without the machinery that connects a nuclear cross section to a count in a tank of liquid.



Read left to right this is a prediction. Read right to left it is Part I. Each box below is one subsection, and each introduces exactly the equations needed to cross one arrow.

II.1 Cross Sections and the Astrophysical S-Factor

Why $\sigma(E)$ Is the Wrong Variable

At stellar energies the cross section for a charged-particle reaction varies by tens of orders of magnitude across a few keV. Almost all of that variation comes from two effects that have nothing to do with nuclear structure.

1. Geometry. The reaction cross section scales with the de Broglie area:

$$\sigma \;\propto\; \pi\bar{\lambda}^2 = \pi\left(\frac{\hbar}{p}\right)^2 \;\propto\; \frac{1}{E}$$

2. The Coulomb barrier. Classically the reaction is forbidden: at $T_c \approx 1.5 \times 10^7$ K the thermal energy is $kT \approx 1.3$ keV, while the barrier for $p + p$ at nuclear contact is ~ 550 keV. Fusion happens only by tunnelling, and the WKB transmission probability is

$$P \;\simeq\; \exp(-2\pi\eta), \qquad \eta = \frac{Z_1 Z_2 e^2}{\hbar v}$$

where η is the Sommerfeld parameter. In practical units,

$$2\pi\eta = 31.29\, Z_1 Z_2 \left(\frac{\mu}{E}\right)^{1/2}, \qquad E \text{ in keV, } \mu \text{ in amu}$$

with $\mu = A_1 A_2/(A_1 + A_2)$ the reduced mass.

Defining the S-Factor

Divide out both trivial effects. What remains is, by construction, whatever the nuclear physics is actually doing:

$$\boxed{\sigma(E) \;=\; \frac{S(E)}{E}\,\exp(-2\pi\eta)} \qquad \Longleftrightarrow \qquad S(E) = E\,\sigma(E)\,e^{2\pi\eta}$$

Why this is the right thing to tabulate

For a non-resonant reaction $S(E)$ is a smooth, slowly varying function — often nearly constant, well represented by $S(E) \approx S(0) + S'(0)E + \frac{1}{2}S''(0)E^2$. That is what makes extrapolation from laboratory energies down to stellar energies even conceivable.

The extrapolation problem

Laboratory measurements are typically possible only at $E \gtrsim 100$ keV, because below that the cross section falls below any achievable count rate. Solar burning happens near $E \approx 5$–30 keV. Every stellar reaction rate in this course rests on an extrapolation of one or two orders of magnitude in energy, and the uncertainty in that extrapolation is the dominant theoretical uncertainty in solar neutrino predictions.

This is exactly why **LUNA** was built underground at Gran Sasso: to push direct measurements down into the real stellar energy range. Same mountain as Borexino, opposite end of the inference chain.

Representative S-Factors

Reaction	$S(0)$	Comment
$p(p, e^+\nu)d$	4.01×10^{-25} MeV b	weak interaction
$d(p, \gamma)^3\mathrm{He}$	$\sim 2.5 \times 10^{-7}$ MeV b	electromagnetic
$^3\mathrm{He}(^3\mathrm{He}, 2p)^4\mathrm{He}$	5.21 MeV b	strong interaction
$^3\mathrm{He}(\alpha, \gamma)^7\mathrm{Be}$	0.56 keV b	sets pp-II/III branching
$^7\mathrm{Be}(p, \gamma)^8\mathrm{B}$	20.8 eV b	sets the $^8\mathrm{B}$ flux
$^{12}\mathrm{C}(p, \gamma)^{13}\mathrm{N}$	1.34 keV b	CNO entry point
$^{14}\mathrm{N}(p, \gamma)^{15}\mathrm{O}$	1.66 keV b	CNO bottleneck

- Note the range: twenty-five orders of magnitude between $p + p$ and $^3\mathrm{He} + {}^3\mathrm{He}$. That gap is not a barrier effect — the barrier has already been divided out. It is the difference between the weak and the strong interaction, laid bare.
- This is the quantitative version of the statement in Part I that the pp chain is throttled by its first step. $S(0)$ makes it a number rather than an assertion.
- Note also that $^{14}\mathrm{N}(p, \gamma)^{15}\mathrm{O}$ does not have an anomalously small $S(0)$. Its slowness comes from the *barrier* ($Z_1 Z_2 = 7$), which the next subsection quantifies.

II.2 Thermonuclear Reaction Rates and the Gamow Peak

The Reaction Rate per Unit Volume

For two species with number densities n_1 and n_2:

$$\boxed{r_{12} = \frac{n_1 n_2}{1 + \delta_{12}} \langle \sigma v \rangle_{12}} \qquad \left[\mathrm{cm}^{-3}\,\mathrm{s}^{-1}\right]$$

The Kronecker δ_{12} prevents double-counting identical pairs — it is why factors of $\frac{1}{2}$ appear in the $p + p$ and ${}^3\mathrm{He} + {}^3\mathrm{He}$ terms of the network equations later.

In a stellar plasma the relative velocities follow Maxwell–Boltzmann, so the rate coefficient is the average

$$\langle \sigma v \rangle = \left(\frac{8}{\pi \mu}\right)^{1/2} \frac{1}{(kT)^{3/2}} \int_0^\infty \sigma(E)\, E\, e^{-E/kT}\, dE$$

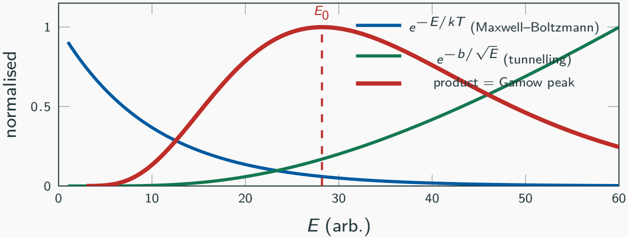
Substituting the S-factor definition gives the central integral of nuclear astrophysics:

$$\boxed{\langle \sigma v \rangle = \left(\frac{8}{\pi \mu}\right)^{1/2} \frac{1}{(kT)^{3/2}} \int_0^\infty S(E)\, \exp\left[-\frac{E}{kT} - \frac{b}{\sqrt{E}}\right] dE}$$

with $b = 2\pi\eta\sqrt{E} = 0.989\, Z_1 Z_2 \sqrt{\mu}$ MeV$^{1/2}$, a constant for a given reaction.

The Gamow Peak

The integrand is the product of two competing exponentials.



Setting $\frac{d}{dE}\left[\frac{E}{kT}+\frac{b}{\sqrt{E}}\right]=0$ locates the peak, and expanding the exponent to second order about it gives a Gaussian of width ΔE_0:

$$E_0=\left(\frac{b\,kT}{2}\right)^{2/3}=1.22\left(Z_1^2 Z_2^2 \mu\, T_6^2\right)^{1/3}\ \text{keV},\qquad \Delta E_0=4\left(\frac{E_0\,kT}{3}\right)^{1/2}=0.749\left(Z_1^2 Z_2^2 \mu\, T_6^5\right)^{1/6}\ \text{keV}$$

Evaluating the Integral

Treating $S(E) \approx S(E_0)$ across the narrow peak, the integral becomes elementary:

$$\boxed{\langle \sigma v \rangle \simeq \left(\frac{2}{\mu}\right)^{1/2} \frac{\Delta E_0}{(kT)^{3/2}} S(E_0)\, e^{-\tau}, \qquad \tau \equiv \frac{3E_0}{kT} = 42.46 \left(\frac{Z_1^2 Z_2^2 \mu}{T_6}\right)^{1/3}}$$

The whole temperature dependence now sits in a single dimensionless number τ. Differentiating,

$$n \equiv \frac{d \ln\langle \sigma v \rangle}{d \ln T} = \frac{\tau - 2}{3} \qquad \Longrightarrow \qquad \varepsilon \propto \rho X_1 X_2 T^n$$

This is the origin of every exponent quoted in Part I

The famous T^4, T^{17} and T^{40} behaviours are not empirical fits. They are $\frac{1}{3}(\tau - 2)$ evaluated at the relevant temperature, and τ depends on the charges only through $(Z_1 Z_2)^{2/3}$.

The power law is a *local* approximation: n itself drifts with temperature, because $\tau \propto T^{-1/3}$.

Worked Numbers at the Solar Centre

Take $T_6 = 15$, so $kT = 1.29$ keV. Students should be able to reproduce every entry in this table with a calculator.

Reaction	$Z_1 Z_2$	μ (amu)	E_0 (keV)	τ	$n = (\tau - 2)/3$
$p + p$	1	0.50	5.9	13.7	3.9
$p + {}^{12}\mathrm{C}$	6	0.92	24.4	56.7	18.2
$p + {}^{14}\mathrm{N}$	7	0.93	26.5	61.6	19.9
${}^{3}\mathrm{He} + {}^{3}\mathrm{He}$	4	1.50	22.0	51.2	16.4

- $E_0 \approx 6$ keV for $p + p$ against $kT = 1.29$ keV: fusion occurs at four to five times the mean thermal energy, far out in the tail of the Maxwell distribution, yet still far below the barrier. Both statements matter.
- $p + {}^{14}\mathrm{N}$ gives $n \approx 20$, against $n \approx 4$ for $p + p$. That single contrast is the reason the CNO cycle switches on only in hotter cores, and hence the reason the Sun is a $\sim 1\%$ CNO star while a $2\,M_\odot$ star is a CNO star outright.
- Note E_0 rises with T as $T^{2/3}$: hotter stars burn at higher energies, where $S(E)$ may no longer be the extrapolated $S(0)$.

Electron Screening

The formulae above treat the nuclei as bare. In a plasma they are not: the electron cloud partially shields the repulsion, lowering the effective barrier and *increasing* the rate.

In the weak-screening (Debye–Hückel) limit, appropriate for the solar interior,

$$f_{\mathrm{scr}} = \frac{\langle \sigma v \rangle_{\mathrm{screened}}}{\langle \sigma v \rangle_{\mathrm{bare}}} \simeq \exp\left(\frac{U_e}{kT}\right), \qquad U_e = \frac{Z_1 Z_2 e^2}{R_D}, \qquad R_D = \left(\frac{kT}{4\pi e^2 \rho N_A \zeta}\right)^{1/2}$$

with R_D the Debye radius and ζ a composition-dependent factor.

- In the Sun the enhancement is a few percent for $p + p$, somewhat larger for higher-Z reactions.
- A few percent sounds ignorable — until you recall $\Phi(^{8}\mathrm{B}) \propto T_c^{24}$ and that $^{7}\mathrm{Be}$ has been measured to 2.7%. At current precision, screening is not a correction one may drop.
- In the *laboratory*, screening by bound atomic electrons is a serious systematic in low-energy measurements, and must be removed before an $S(E)$ is quoted. Different screening, same physics, opposite sign of concern.

II.3 Resonant Reactions

When $S(E)$ Is Not Smooth

Everything so far assumed non-resonant behaviour. If the compound nucleus has a state at an energy accessible in the entrance channel, the cross section develops a sharp peak described by the Breit–Wigner formula:

$$\boxed{\sigma_{BW}(E) = \pi \bar{\lambda}^2\, \omega\, \frac{\Gamma_a \Gamma_b}{(E - E_r)^2 + (\Gamma/2)^2}}$$

- E_r: resonance energy; Γ_a, Γ_b: partial widths for entrance and exit channels; Γ: total width.
- $\omega = \dfrac{2J+1}{(2j_1+1)(2j_2+1)}$ is the statistical spin factor, and it is precisely here that the **shell model** enters: it supplies J^π for the resonant state, and the selection rules determine which Γ_b are non-zero.

For a *narrow* resonance ($\Gamma \ll E_r$) the Maxwellian average collapses to a single term:

$$\langle \sigma v \rangle = \left(\frac{2\pi}{\mu k T} \right)^{3/2} \hbar^2\, (\omega\gamma)\, e^{-E_r/kT}, \qquad \omega\gamma \equiv \omega\, \frac{\Gamma_a \Gamma_b}{\Gamma}$$

The entire rate is governed by one number, the resonance strength $\omega\gamma$, and one exponential.

Two Consequences Worth Drawing Out

1. Resonances make rates violently temperature-sensitive. The factor $e^{-E_r/kT}$ has logarithmic derivative E_r/kT, which for a resonance a few hundred keV above threshold at $T_8 \sim 1$ is enormous. This, not the three-body nature alone, is where the triple-α T^{40} comes from.

2. A missing resonance can make a reaction impossible — and a required reaction can predict a resonance.

The triple-α process must proceed through an unbound intermediate:

$$^{4}\mathrm{He} + {}^{4}\mathrm{He} \rightleftharpoons {}^{8}\mathrm{Be} \quad \text{(unbound by 92 keV, } \tau \sim 10^{-16}\text{ s)}, \qquad {}^{8}\mathrm{Be} + {}^{4}\mathrm{He} \to {}^{12}\mathrm{C}$$

Computed non-resonantly, the rate cannot produce the observed carbon. Hoyle's 1953 argument was therefore observational: carbon exists, so a 0^+ state must sit just above the 3α threshold at 7.275 MeV. It was found at 7.654 MeV, a mere 0.38 MeV above — comparable to the thermal energy at $T_8 \approx 1$.

The methodological point

This is the mirror image of Part I. There, a measurement in a detector constrained nuclear physics. Here, an astronomical fact — the existence of carbon — predicted a nuclear energy level, which was then found in a laboratory.

II.4 Reaction Networks and the pp Chain

The General Network Equation

A burning region is described by coupled first-order ODEs for the abundances. Using $Y_i = X_i/A_i$ (moles per gram), the general form is

$$\boxed{\frac{dY_i}{dt} = \sum_j \lambda_j Y_j + \sum_{j,k} \rho \frac{Y_j Y_k}{1+\delta_{jk}} N_A \langle \sigma v \rangle_{jk} - \text{(destruction terms)}}$$

where λ_j are decay constants for β decays and electron captures, and $N_A \langle \sigma v \rangle$ is the tabulated rate coefficient from II.2–II.3.

Two derived quantities do most of the work in practice:

lifetime of i against j: $\tau_i(j) = \dfrac{1}{n_j \langle \sigma v \rangle_{ij}}$ energy generation: $\varepsilon = \dfrac{1}{\rho} \sum_{\text{reactions}} r\, Q_{\text{eff}}$

The strategy that makes networks tractable

Compare lifetimes. If $\tau_i(j) \ll \tau_i(k)$, channel k can be dropped entirely. Almost every simplification in this course is an application of that one comparison — and the deciding factor is usually abundance, not Q-value.

The pp Chain Network

Writing H, D, ${}^3\mathrm{He}$, ${}^4\mathrm{He}$, ${}^7\mathrm{Be}$, ${}^7\mathrm{Li}$ for number densities and λ_{ij} for rate coefficients:

$$\begin{aligned}
\frac{d(H)}{dt} &= -2\lambda_{11}\frac{H^2}{2} - \lambda_{12}HD + 2\lambda_{33}\frac{({}^3\mathrm{He})^2}{2} - \lambda_{17}H({}^7\mathrm{Be}) - \lambda_{17}^{*}H({}^7\mathrm{Li}) \\
\frac{d(D)}{dt} &= \lambda_{11}\frac{H^2}{2} - \lambda_{12}HD \\
\frac{d({}^3\mathrm{He})}{dt} &= \lambda_{12}HD - 2\lambda_{33}\frac{({}^3\mathrm{He})^2}{2} - \lambda_{34}({}^3\mathrm{He})({}^4\mathrm{He}) \\
\frac{d({}^4\mathrm{He})}{dt} &= \lambda_{33}\frac{({}^3\mathrm{He})^2}{2} - \lambda_{34}({}^3\mathrm{He})({}^4\mathrm{He}) + 2\lambda_{17}H({}^7\mathrm{Be}) + 2\lambda_{17}^{*}H({}^7\mathrm{Li}) \\
\frac{d({}^7\mathrm{Be})}{dt} &= \lambda_{34}({}^3\mathrm{He})({}^4\mathrm{He}) - \lambda_{e7}n_e({}^7\mathrm{Be}) - \lambda_{17}H({}^7\mathrm{Be}) \\
\frac{d({}^7\mathrm{Li})}{dt} &= \lambda_{e7}n_e({}^7\mathrm{Be}) - \lambda_{17}^{*}H({}^7\mathrm{Li})
\end{aligned}$$

Minus signs are destruction, plus signs production; the $\frac{1}{2}$ factors are the δ_{jk} of the previous slide, and the leading 2's count nuclei consumed or produced per reaction. Every branching ratio measured in Section I.2 is a ratio of terms in these six equations.

Quasi-Equilibrium: Deuterium

Production is weak (slow), destruction is electromagnetic (fast), so D reaches a steady state almost immediately. Setting $dD/dt = 0$:

$$\boxed{\left(\frac{D}{H}\right)_e = \frac{\langle\sigma v\rangle_{pp}}{2\langle\sigma v\rangle_{pd}}} \qquad \Longrightarrow \qquad \left(\frac{D}{H}\right)_e \approx 10^{-18} \text{ at } T_6 = 5$$

The corresponding lifetimes state the hierarchy directly:

$$\tau_H(p) \approx 10^{10} \text{ yr}, \qquad \tau_d(p) \approx 1.6 \text{ s}$$

The observed value is $(D/H)_{\rm obs} \approx 10^{-5}$

Thirteen orders of magnitude too high, and no adjustment of stellar physics can close that gap — stars *destroy* deuterium and never accumulate it.

Inference: The deuterium we observe predates stellar burning. It is a relic of Big Bang nucleosynthesis. A failed prediction, taken seriously rather than patched, became a pillar of cosmology.

Quasi-Equilibrium: $^3\mathrm{He}$ and the Branch Point

The same procedure applied to $^3\mathrm{He}$, keeping only the dominant destruction channel $^3\mathrm{He} + {}^3\mathrm{He}$:

$$\frac{d(^3\mathrm{He})}{dt} = \frac{H^2 \langle \sigma v \rangle_{11}}{2} - (^3\mathrm{He})^2 \langle \sigma v \rangle_{33} \qquad \Longrightarrow \qquad \boxed{\left(\frac{^3\mathrm{He}}{H}\right)_e = \left(\frac{\langle \sigma v \rangle_{11}}{2\langle \sigma v \rangle_{33}}\right)^{1/2}}$$

Note the square root: because $^3\mathrm{He}$ destroys itself two at a time, its equilibrium abundance scales as the *half* power of the rate ratio, unlike deuterium.

Time to reach equilibrium (at $X_H = X_{He} = 0.5$, $\rho = 100$ g cm^{-3}):

T_6	t_f (99% of equilibrium)	Consequence
5	$\sim 10^{25}$ yr	never, for any star
15	$\sim 10^{6}$ yr	comparable to early evolutionary timescales
30	$\sim 10^{5}$ yr	rapid

Inference: The $^3\mathrm{He}$ abundance depends on stellar age and on local temperature, so it varies with radius. That radial gradient is later dredged to the surface — and it is why the measured pp-I / pp-II branching ratio ($R_{\mathrm{I/II}} = 0.178$ in Section I.2) is a genuine test of this equation rather than a triviality.

Why $^4\mathrm{Li}$ Does Not Save Time, and What pp-I Costs

Students always ask why abundant protons do not finish the job via $^3\mathrm{He}(p,\gamma)^4\mathrm{Li}$. Two reasons, either sufficient:

- $Q = -2.5$ MeV — the reaction is *endothermic*.
- $^4\mathrm{Li}$ is not particle-stable; it does not survive long enough to β-decay.

So pp-I terminates on $^3\mathrm{He} + {}^3\mathrm{He}$, and the energy bookkeeping runs:

$$3p \to {}^3\mathrm{He}: \quad Q = 6.936 \text{ MeV}, \quad \langle E_\nu \rangle = 0.263 \text{ MeV} \quad \Rightarrow \quad E_{\mathrm{eff}} = 6.673 \text{ MeV}$$
$$^3\mathrm{He} + {}^3\mathrm{He} \to {}^4\mathrm{He} + 2p: \quad Q = 12.860 \text{ MeV}$$

At equilibrium ($2r_{33} = r_{11}$) the total is

$$\boxed{\varepsilon_{pp} = 13.103\, r_{11} \text{ MeV cm}^{-3}\text{s}^{-1}}$$

set entirely by the slow $p + p$ rate, exactly as the $S(0)$ table predicted.

Branch	Share	Q_{eff}	ν loss
pp-I	85%	26.20 MeV	2%
pp-II	15% (with III)	25.66 MeV	4%
pp-III	0.02%	19.17 MeV	28.3%

How Little of the Sun Is Burning

Solving the network under solar conditions gives an energy production rate per gram, which can be compared directly with the observed luminosity:

$$\varepsilon_{\rm tot} = 5.1 \times 10^7 \text{ MeV g}^{-1}\text{s}^{-1}, \qquad L_\odot = 2.4 \times 10^{39} \text{ MeV s}^{-1}$$

$$m_\odot^{\rm burning} = \frac{L_\odot}{\varepsilon_{\rm tot}} = 4.7 \times 10^{31} \text{ g} \qquad \text{against} \qquad M_\odot = 2 \times 10^{33} \text{ g}$$

$$\boxed{\text{Only} \approx 2.4\% \text{ of the Sun's mass is actively burning hydrogen}}$$

Inference: Fusion is confined to a small, extremely temperature-sensitive central region. A star does not die when it runs out of hydrogen — it dies when it runs out of hydrogen in the small central region hot enough to burn it. Shell burning, red giants and dredge-up all follow from this one fact.

II.5 The CNO Cycle: Equilibrium and Energy Generation

The Cycle as a Set of Rate Equations

For sub-cycle I, with $\lambda_i = \rho X_H N_A \langle \sigma v \rangle_i$ the proton-capture rate on species i and λ^β the decay constants:

$$\frac{dY(^{12}\mathrm{C})}{dt} = \lambda_{15}\, Y(^{15}\mathrm{N}) - \lambda_{12}\, Y(^{12}\mathrm{C})$$

$$\frac{dY(^{13}\mathrm{N})}{dt} = \lambda_{12}\, Y(^{12}\mathrm{C}) - \lambda^\beta_{13}\, Y(^{13}\mathrm{N})$$

$$\frac{dY(^{13}\mathrm{C})}{dt} = \lambda^\beta_{13}\, Y(^{13}\mathrm{N}) - \lambda_{13}\, Y(^{13}\mathrm{C})$$

$$\frac{dY(^{14}\mathrm{N})}{dt} = \lambda_{13}\, Y(^{13}\mathrm{C}) - \lambda_{14}\, Y(^{14}\mathrm{N})$$

$$\frac{dY(^{15}\mathrm{O})}{dt} = \lambda_{14}\, Y(^{14}\mathrm{N}) - \lambda^\beta_{15}\, Y(^{15}\mathrm{O})$$

$$\frac{dY(^{15}\mathrm{N})}{dt} = \lambda^\beta_{15}\, Y(^{15}\mathrm{O}) - \lambda_{15}\, Y(^{15}\mathrm{N})$$

Summing all six lines gives $\frac{d}{dt} \sum_i Y_i = 0$: the total number of catalysts is conserved. That is the mathematical statement of catalysis, and it is why Φ_{CNO} measures the *sum* $N_{\mathrm{C+N}}$ rather than any individual isotope.

Equilibrium Abundances and the Bottleneck

In steady state every term above vanishes, so all the fluxes are equal:

$$\lambda_{12}\, Y(^{12}\mathrm{C}) = \lambda_{13}\, Y(^{13}\mathrm{C}) = \lambda_{14}\, Y(^{14}\mathrm{N}) = \lambda_{15}\, Y(^{15}\mathrm{N}) \;\equiv\; F$$

$$\boxed{Y_i \;\propto\; \frac{1}{\lambda_i} \;\propto\; \frac{1}{N_A\langle\sigma v\rangle_i}}$$

The slowest reaction accumulates the most nuclei

The species waiting at the slowest step piles up until its abundance compensates for its small rate coefficient. Since $^{14}\mathrm{N}(p,\gamma)^{15}\mathrm{O}$ is by far the slowest, nitrogen-14 accumulates and carbon is depleted.

Ratio	Initial (ISM / solar)	CNO equilibrium
$^{12}\mathrm{C}/^{13}\mathrm{C}$	≈ 89	≈ 3.5
$^{14}\mathrm{N}/^{12}\mathrm{C}$	≈ 0.25	$\gtrsim 20$
$\mathrm{C}+\mathrm{N}+\mathrm{O}$	—	unchanged

Red giants after first dredge-up show $^{12}\mathrm{C}/^{13}\mathrm{C} \approx 20$: processed material diluted by an unprocessed envelope, where the *degree* of dilution measures how deep that envelope reached.

CNO Energy Generation and Neutrino Production

Because every step must run at the common rate F, the energy generation rate is simply that flux times the cycle Q-value:

$$\varepsilon_{\mathrm{CNO}} = \frac{F\,Q_{\mathrm{eff}}}{\rho}, \qquad \varepsilon_{\mathrm{CNO}} \;\propto\; \rho\, X_H\, X_{\mathrm{CN}}\, T^{\,n}, \qquad n \approx 20 \text{ near } T_6 = 15$$

The linear dependence on X_{CN} is the entire basis of the Borexino metallicity measurement — and, unlike the T^n factor, it cannot be mimicked by any other flux.

Neutrino production follows immediately. Since each turn of the cycle passes through exactly one $^{13}\mathrm{N}$ and one $^{15}\mathrm{O}$ decay,

$$\Phi(^{13}\mathrm{N}) \;\simeq\; \Phi(^{15}\mathrm{O}) \;\simeq\; F \times (\text{geometry}), \qquad \Phi(^{17}\mathrm{F}) \ll \Phi(^{13}\mathrm{N})$$

Why Borexino fitted the three CNO components as one

The individual fluxes are *not* independent parameters. Their ratios are fixed by the cycle equations above and by the sub-cycle branching, so the analysis constrained the ratio from the solar model and fitted a single overall normalisation. This is a theoretical input to an experimental result, and students should be able to say exactly where it came from — it came from the steady-state condition on this slide.

II.6 From a Core Reaction to a Count in a Detector

Step 1 — The Shape of the Emitted Neutrino Spectrum

Three-body β^+ decay $\Rightarrow$ a continuum. For an allowed transition, the statistical (phase-space) factor gives

$$N(E_\nu)\, dE_\nu \;\propto\; F(Z, E_e)\, p_e E_e\, E_\nu^2\, dE_\nu\,, \qquad E_e = Q - E_\nu$$

with $F(Z, E_e)$ the Fermi function correcting for Coulomb distortion of the outgoing positron. This produces the smooth $^{13}\mathrm{N}$, $^{15}\mathrm{O}$, $^{17}\mathrm{F}$, *pp* and $^{8}\mathrm{B}$ spectra of Part I.

Two-body electron capture $\Rightarrow$ a line. With only two products, energy conservation fixes the neutrino energy exactly:

$$E_\nu = Q_{EC} - E_x$$

Hence $^{7}\mathrm{Be}$ gives two lines — 862 keV (90%, ground state) and 384 keV (10%, first excited state of $^{7}\mathrm{Li}$) — and *pep* gives a single line at 1.44 MeV.

Inference: The continuum-versus-line distinction is pure kinematics, decided by the number of bodies in the final state. It is what allowed Borexino to identify components by shape alone in a fit with no event-by-event tagging.

Step 2 — The Luminosity Constraint

The fluxes are not free. If the Sun is in steady state, every $^4\mathrm{He}$ built releases the same total energy, part of which escapes as neutrinos:

$$\boxed{\frac{L_\odot}{4\pi d^2} = \sum_i \left(\frac{Q}{2} - \langle E_\nu\rangle_i\right)\Phi_i} \qquad Q = 26.73\ \mathrm{MeV}$$

The sum runs over all neutrino-producing branches, d is the Earth–Sun distance, and $\langle E_\nu\rangle_i$ is the mean neutrino energy of branch i.

- Reading it forwards: it predicts the total flux $\approx 6\times10^{10}\ \mathrm{cm^{-2}s^{-1}}$ from the measured luminosity alone, with essentially no nuclear physics. This is Problem 1 of Part I.
- Reading it backwards: it is the constraint that fixed the *pep* rate to 1.4% in the CNO analysis. Because $\langle E_\nu\rangle_{\mathrm{CNO}}$ contributes so weakly to the sum, the constraint is *insensitive* to the CNO rate — which is precisely why using it is not circular.
- Reading it as a test: comparing L_{nuc} with L_{phot} probes thermal equilibrium over the photon diffusion time, as in Section I.2.

Step 3 — Flavour Conversion

Neutrinos are produced as ν_e but detected as a mixture. In the adiabatic MSW-LMA approximation the survival probability is

$$\boxed{P_{ee}(E_\nu) = \frac{1}{2} + \frac{1}{2}\cos 2\theta_m^{\mathrm{prod}}\,\cos 2\theta_{12}}$$

where θ_{12} is the vacuum mixing angle and θ_m^{prod} the effective mixing angle *at the production point*, given by

$$\cos 2\theta_m = \frac{\cos 2\theta_{12} - \beta}{\sqrt{(\cos 2\theta_{12} - \beta)^2 + \sin^2 2\theta_{12}}}, \qquad \beta = \frac{2\sqrt{2}\,G_F\,n_e\,E_\nu}{\Delta m^2_{21}}$$

- $\beta \ll 1$ (low E_ν): matter is irrelevant, $P_{ee} \to 1 - \frac{1}{2}\sin^2 2\theta_{12} \approx 0.55$. *the pp, $^7\mathrm{Be}$ regime*
- $\beta \gg 1$ (high E_ν): $\cos 2\theta_m \to -1$ and $P_{ee} \to \sin^2\theta_{12} \approx 0.31$. *the $^8\mathrm{B}$ regime*
- The transition between the two is the downward step plotted in Section I.2 — and note it depends on n_e at the production point, i.e. on the solar model.

Step 4 — The Detection Cross Section and the Count Rate

Borexino observes $\nu + e^- \to \nu + e^-$. The differential cross section in electron recoil kinetic energy T is

$$\frac{d\sigma}{dT} = \frac{2G_F^2 m_e}{\pi}\left[g_L^2 + g_R^2\left(1 - \frac{T}{E_\nu}\right)^2 - g_L g_R \frac{m_e T}{E_\nu^2}\right]$$

with $g_L = \frac{1}{2} + \sin^2\theta_W$ for ν_e but $g_L = -\frac{1}{2} + \sin^2\theta_W$ for $\nu_{\mu,\tau}$, since only ν_e has the charged-current contribution.

Why this matters for interpreting Part I

$\sigma(\nu_e e) \approx 6\,\sigma(\nu_{\mu\tau} e)$. Borexino is therefore *not* blind to the oscillated neutrinos — it sees them at roughly one sixth sensitivity. Every quoted flux already accounts for this.

Assembling everything, the predicted count rate is

$$\boxed{R = N_e \int dE_\nu\, \Phi(E_\nu) \int_0^{T_{\max}} dT \left[P_{ee}\frac{d\sigma_{\nu_e}}{dT} + (1 - P_{ee})\,\frac{d\sigma_{\nu_{\mu\tau}}}{dT}\right] \mathcal{R}(T, T')}$$

with N_e the number of target electrons and $\mathcal{R}$ the detector response function. This single expression is the bridge between Part II and every number in Part I.

II.7 The Standard Solar Model

The Four Structure Equations

Everything above assumed a known $\rho(r)$, $T(r)$ and composition. Those come from solving, for a spherically symmetric star in hydrostatic equilibrium:

$$\frac{dP}{dr} = -\frac{Gm(r)\rho}{r^2} \qquad \text{hydrostatic equilibrium} \qquad\qquad \frac{dL}{dr} = 4\pi r^2 \rho\, \varepsilon(\rho, T, X_i) \qquad \text{energy generation}$$

$$\frac{dm}{dr} = 4\pi r^2 \rho \qquad \text{mass conservation} \qquad\qquad \frac{dT}{dr} = -\frac{3\kappa\rho L}{16\pi a c T^3 r^2} \qquad \text{radiative transport}$$

(with the radiative gradient replaced by the adiabatic one wherever the Schwarzschild criterion signals convection).

Where the previous six subsections plug in

The term $\varepsilon(\rho, T, X_i)$ in the third equation is exactly what II.1–II.5 computed. The structure equations and the reaction network are solved *together*: composition changes alter ε and κ, which alter $T(r)$, which alters ε again. That coupling is why a change in metallicity propagates into every neutrino flux at once.

Calibration, and Why $\Phi(^8\mathrm{B}) \propto T_c^{24}$

Input physics required: equation of state, radiative opacities $\kappa(\rho, T, X_i)$, nuclear reaction rates (II.1–II.3), and element diffusion.

Calibration: three free parameters — initial helium Y_i, initial $(Z/X)_i$, and the mixing-length parameter — are tuned until the model at $t = 4.57$ Gyr reproduces the observed $L_\odot$, $R_\odot$ and surface $(Z/X)_\odot$. The neutrino fluxes are then predictions, not fits.

Where the steep sensitivities come from. Combine two facts:

1. From II.2, each branch has $\varepsilon_i \propto T^{n_i}$ with n_i set by $\tau = 42.46(Z_1^2 Z_2^2 \mu / T_6)^{1/3}$.
2. The luminosity is *fixed* by observation, so raising T_c cannot raise the total energy output — it can only redistribute it between branches.

Hence the fluxes of the rare, high-$Z_1 Z_2$ branches respond enormously while $\Phi(pp)$ barely moves:

$$\Phi(pp) \propto T_c^{-1.1}, \qquad \Phi(^7\mathrm{Be}) \propto T_c^{+11}, \qquad \Phi(^8\mathrm{B}) \propto T_c^{+24}, \qquad \Phi(\mathrm{CNO}) \propto T_c^{+20} X_{\mathrm{CN}}$$

The negative exponent for pp is not a typo: it is the signature of a branch that must *give up* share as the others become more efficient.

The Metallicity Problem, Quantitatively

Now the chain of Part I can be written as a sequence of dependencies rather than an assertion:

$$Z \uparrow \Longrightarrow \kappa \uparrow \Longrightarrow \left|\frac{dT}{dr}\right| \uparrow \Longrightarrow T_c \uparrow \Longrightarrow \Phi(^7\mathrm{Be}),\ \Phi(^8\mathrm{B}),\ \Phi(\mathrm{CNO}) \uparrow\uparrow$$

and, for CNO only, an *additional* direct factor:

$$\Phi(\mathrm{CNO}) \propto X_{\mathrm{CN}} \times T_c^{20}$$

Why CNO is the decisive measurement

Every other flux constrains the *product* of composition and temperature effects, and cannot separate them. CNO carries the composition dependence twice — once through opacity and once directly, as a catalyst — which is what makes it able to break the degeneracy.

Turning the measured rate into an abundance requires the CNO reaction rates from II.1–II.3 and the core temperature from this subsection. This is the "weakest link" flagged in the Part I inference chain, and it is the reason the final answer is quoted as $N_{\mathrm{CN}} = 5.81\,^{+1.22}_{-0.94} \times 10^{-4}$ with an asymmetric error rather than a clean number.

II.8 Consolidation

The Master Table

Quantity	Expression	What it explains in Part I
S-factor	$\sigma(E) = \frac{S(E)}{E} e^{-2\pi\eta}$	$S_{pp} \sim 10^{-25}$ vs $S_{33} \sim 1$: why the pp chain is slow
Rate	$r = \frac{n_1 n_2}{1+\delta_{12}} \langle \sigma v \rangle$	the $\frac{1}{2}$ factors in the network equations
Gamow peak	$E_0 = 1.22(Z_1^2 Z_2^2 \mu T_6^2)^{1/3}$ keV	$E_0 \approx 6$ keV for pp: burning far out in the Maxwell tail
Temperature index	$n = (\tau - 2)/3$, $\tau = 42.46(Z_1^2 Z_2^2 \mu / T_6)^{1/3}$	T^4 vs T^{20}: why the Sun is only a 1% CNO star
Narrow resonance	$\langle \sigma v \rangle \propto (\omega\gamma) e^{-E_r/kT}$	the Hoyle state; $^{14}\mathrm{N}(p, \gamma)$ resonances
Network	$dY_i/dt = \sum \lambda Y + \sum \rho YYN_A \langle \sigma v \rangle$	branching ratios; $R_{\mathrm{I/II}} = 0.178$
Cycle equilibrium	$Y_i \propto 1/\langle \sigma v \rangle_i$	$^{14}\mathrm{N}$ accumulates; $^{12}\mathrm{C}/^{13}\mathrm{C}$ drops to ~ 20 in giants
Catalyst conservation	$\frac{d}{dt} \sum_i Y_i = 0$	why Φ_{CNO} measures $N_{\mathrm{C+N}}$, not one isotope
Luminosity constraint	$\frac{L_\odot}{4\pi d^2} = \sum_i (\frac{Q}{2} - \langle E_\nu \rangle_i) \Phi_i$	the pep constraint; $L_{\mathrm{nuc}} = L_{\mathrm{phot}}$; total flux estimate
Survival probability	$P_{ee} = \frac{1}{2} + \frac{1}{2} \cos 2\theta_m \cos 2\theta_{12}$	the MSW step from 0.55 to 0.35
Count rate	$R = N_e \int dE_\nu \Phi \int dT \frac{d\sigma}{dT} \mathcal{R}$	converts every flux in Part I into cpd/100 t
Flux sensitivity	$\Phi(^8\mathrm{B}) \propto T_c^{24}$, $\Phi_{\mathrm{CNO}} \propto X_{\mathrm{CN}} T_c^{20}$	why CNO alone can weigh the core

Summary of Part II

- **Cross sections** are dominated by geometry and Coulomb tunnelling. Dividing both out leaves the *S*-factor, which is smooth and can be extrapolated — and whose extrapolation is the dominant theoretical uncertainty in every solar neutrino prediction.
- **Reaction rates** follow from a Maxwellian average whose integrand peaks in the Gamow window at $E_0 \gg kT$ but $\ll$ the barrier. A single parameter τ controls the temperature dependence, giving $n = (\tau - 2)/3$ and hence T^4, T^{20}, T^{40}.
- **Resonances** replace the smooth $S(E)$ by a Breit–Wigner peak and concentrate the entire rate into $(\omega\gamma)e^{-E_r/kT}$. The shell model supplies J^π; without a resonance, carbon could not exist.
- **Networks** of coupled ODEs, simplified by comparing lifetimes, give quasi-equilibrium abundances — and one of them, $(D/H)_e \sim 10^{-18}$, fails against observation by thirteen orders of magnitude, which is how we know deuterium is cosmological.
- **Catalytic cycles** conserve $\sum_i Y_i$ and drive $Y_i \propto 1/\langle\sigma v\rangle_i$, so the slowest step accumulates. This gives the nitrogen enrichment, the low $^{12}\mathrm{C}/^{13}\mathrm{C}$, and the linear dependence of Φ_{CNO} on X_{CN} that Borexino exploited.
- **The bridge to the detector** is the luminosity constraint, the MSW survival probability and the ν–e elastic cross section — three well-determined pieces of physics that convert a core reaction rate into counts per day per hundred tonnes.

Problems and Discussion — Part II

1. Compute E_0, ΔE_0 and τ for ${}^3\mathrm{He} + {}^4\mathrm{He} \to {}^7\mathrm{Be}$ at $T_6 = 15$. Compare n with the value for ${}^3\mathrm{He} + {}^3\mathrm{He}$ and explain the resulting temperature dependence of the pp-I / pp-II branching ratio.
2. $S_{pp}(0)$ is 25 orders of magnitude below $S_{33}(0)$, yet both reactions proceed in the same plasma. Where does the compensating factor come from?
3. Show that $\Phi(pp)$ must have a *negative* temperature exponent, given that $L_\odot$ is fixed. What physical statement does the sign encode?
4. Starting from $Y_i \propto 1/\langle \sigma v \rangle_i$, estimate the equilibrium ${}^{14}\mathrm{N}/{}^{12}\mathrm{C}$ ratio using the $S(0)$ values and τ from II.2. Compare with the quoted $\gtrsim 20$.
5. A resonance is discovered at $E_r = 400$ keV in a reaction burning at $T_8 = 1$. By what factor does the rate change if E_r is remeasured as 380 keV? Comment on the precision required of nuclear experiments.
6. Use the luminosity constraint to estimate the total solar neutrino flux at Earth. Why does the answer barely depend on the branching ratios?
7. Explain, using the P_{ee} formula, why the MSW transition occurs at a few MeV rather than at some other energy, and what would change if n_e in the core were 30% larger.

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End of Part II.

Part I measured the Sun's core.

Part II is why those numbers mean what they mean.